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INSTRUCTIONS TO CANDIDATES

- This group assignment will contribute 50% to your final grade.

Learning Outcome:

CLO3: Prepare a seminar portfolio that documents the student's activities that the students able to explain and analyze different analytics problems in the world of business

IMPORTANT

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Descriptive and Predictive Analytics Using Brazilian Olist E-Commerce Dataset

Abstract

This paper aimed to analyse the Brazilian E-Commerce Public Dataset by Olist through descriptive and predictive analytics in order to support data-driven decision-making. The dataset, which contains information on customer details, orders, products, freight costs and delivery performance, was prepared through data cleaning techniques, merging, transformation and feature engineering. Microsoft Power BI was also used to conduct descriptive analytics such as summary statistics and visualisations to analyse delivery performance, shipping costs, geographical trends, product-category relationships, and anomalies. Predictive analytics methods integrated into this study includes Random Forest, Logistic Regression, and XGBoost in order to predict the order timeliness. To evaluate the performance of machine learning results, evaluation metrics were included such as Accuracy, Balanced Accuracy, Recall, F1-Score, ROC-AUC and PR-AUC. The results showed that XGBoost had the best overall performance among the three classification models with the highest balanced accuracy of 71.06%, recall of 61.97%, F1-Score of 0.2849, ROC-AUC (0.7894), PR-AUC of 0.2520. With a stronger ability to identify late deliveries out of all three machine learning algorithms, XGBoost was selected as the best performing model.

1.0 Introduction

Today, the e-commerce sector is experiencing significant growth in both scale and revenue due to rapid technological advancements. Unlike in the past, when traditional stores relied heavily on manual operations and served a limited customer base, technology has enabled businesses to reach customers on a much larger scale. This has increased sales opportunities and accelerated business growth, particularly for small and medium-sized enterprises (SMEs). Instead of relying solely on customers visiting physical stores, many SMEs can now connect with a wider audience through third-party e-commerce platforms. This rapid growth has also generated massive amounts of data from multiple dimensions, such as logistics, transactions, and customer activities. These data are highly valuable because they often contain hidden insights waiting to be uncovered. Once these insights are identified, companies can take potential actions that improve business performance and support better decision-making. Therefore, data has become a critical asset in the e-commerce sector, especially in Brazil. As e-commerce demand in Brazil continues to grow rapidly, the industry generated approximately US\$36.3 billion in revenue in 2025 (International Trade Administration, 2025).

One of the most well-known e-commerce platforms in Brazil is Olist. The platform helps small and medium sized organization expand their online exposure by connecting them with a large customer base through multiple online marketplaces. Besides, to provide marketplace access, Olist also offers integrated logistics services to support order fulfilment and delivery. However, operating a multivendor marketplace presents significant operational challenges. The company must coordinate thousands of orders from different sellers while ensuring timely delivery, meeting customer expectations, and supporting diverse payment methods. As the number of transactions continues to grow, effectively managing these operations becomes increasingly complex, making data driven decision-making essential for improving efficiency and maintaining customer satisfaction.

To support accurate and effective data driven decision-making, it is essential to transform raw data into useful information that can shape the business strategies. Raw data alone does not solve the operational bottlenecks. It only becomes valuable when it is processed, analysed, and interpreted. Once data is transformed into useful information, Olist can respond to potential issues and make informed decisions. Machine learning can further support this process by analysing historical data to identify

customer behaviour patterns and forecast future trends. By applying machine learning techniques, Olist can predict possible outcomes and prepare appropriate responses in advance. This proactive approach helps reduce potential risks and minimise their impact, as the company can expect challenges before they occur.

In this project, the main objective is to analyse the Olist Brazilian e-commerce dataset by applying exploratory data analysis and machine learning techniques to support data driven decision-making. The project focuses on analysing delivery performance, shipping costs, geographical patterns, product-category patterns, and anomalies to uncover meaningful business insights and improve operational efficiency. In addition, machine learning models, including XGBoost, Random Forest, and Logistic Regression, are applied to predict late deliveries, allowing businesses to identify potential delivery issues and make more informed strategic decisions. Specifically, the project focuses on three main areas. First, product sales analysis is conducted to identify sales trends, popular product categories, and factors influencing product performance. Second, logistics performance is evaluated by analysing delivery days, shipping costs, and delivery performance across product categories to identify potential supply chain challenges. Lastly, customer location-based sales analysis is performed to examine geographical patterns, identify regional differences, and understand how customer distribution influences business performance.

2.0 Literature Review

2.1 Machine Learning for Customer Satisfaction Prediction

A study by Mangla and Soni (2025) proposed a machine learning framework where it predicts the satisfaction of the customers using the Brazilian Olist E-Commerce dataset. Extensive data preprocessing and feature engineering were applied in this study so that variables related to delivery performance, behaviour of customer transaction, payment patterns, characteristics of the products, and temporal information are generated. Random Forest, HistGradientBoosting, XGBoost, and LightGBM were used to evaluate the accuracy, precision, recall, F1-score, and ROC-AUC. Highest accuracy of 82.53% was achieved by the Voting Ensemble model. This demonstrates that the features related to delivery such as the delay in delivery, and carrier time are most influential. This study is important as it utilizes the same Olist dataset and it shows how logistics performance can be optimized, and customer satisfaction can be enhanced by combining descriptive analytics with predictive machine learning techniques.

Similarly, Zagnhoul, Barakat, and Rezk (2024) studied how customer satisfaction can be predicted in e-commerce through the Brazilian Olist dataset. The customer satisfaction was formulated as a binary classification problem. Multiple models such as Logistic Regression, Support Vector Machine, Random Forest, Gradient Boosting Classifier, and a Multi-Layer Perception neural network were compared in this study. Data cleaning, feature engineering, feature selection, standardization, and oversampling were all performed so that the class imbalance can be addressed. Random Forest outperformed the deep learning model by achieving the best result overall after oversampling. The model achieved an accuracy of 92%, AUC-ROC of 0.90, and a weighted F1-score of 0.92. The identified important predictors are delivery time, freight cost payment value, total order value, and geographic factors. This study is relevant as it uses the Olist dataset and demonstrates how machine learning models can support prediction of customer satisfaction and business decisions related to logistics.

Pandey et al. (2023) aims to predict customer satisfaction in Brazilian e-commerce (Olist) using different machine learning models. The models evaluated were LSTM, XGBoost, Support Vector Machine (SVM), Logistic Regression, Random Forest, Naive Bayes and Neural Network. The results showed that LSTM has the highest prediction accuracy (87.09%). XGBoost was the second-best model

with accuracy of 85.36%, followed by Naive Bayes with an accuracy of 84.64%. Besides, the study includes basic descriptive analysis by calculating the average review ratings for different order statuses, plotting the frequency of review scores and examining how the number of items purchased, payment method and delivery cost are related to customer retention. The study mainly focuses on customer satisfaction prediction and model comparison. It does not analyse business performance, logistics trends or customer purchasing behaviour. It also does not perform anomaly detection or provide detailed exploratory data analysis and visualisations for business decision-making.

Gogineni et al. (2026) adopted a broader framework that incorporated customer segmentation, geographical and sentiment analysis using the Brazilian Olist E-Commerce dataset. It develops an end-to-end data analytics and machine learning framework to explore customer purchasing patterns, delivery performance, geographical variations, and customer review sentiment. The study applies RFM (Recency, Frequency, Monetary) modelling and K-Means clustering to segment customers into groups such as high-value loyal customers, at-risk customers, and price-sensitive shoppers. The results show that high-value and loyal customers are mainly concentrated in urban and metropolitan areas. Furthermore, DBSCAN clustering and Moran's I statistic are used to identify geographical patterns related to delivery delays. Natural Language Processing (NLP) techniques, including text preprocessing and sentiment analysis, are applied to extract insights from customer reviews. For predictive modelling, the study trains Logistic Regression, Random Forest, XGBoost, and Support Vector Machine (SVM) models to predict customer review scores. XGBoost achieved the highest accuracy of 92%, indicating strong performance in predicting customer satisfaction.

Wangkiat and Polprasert (2023) examines the results of using three machine learning models, K-Nearest Neighbor (KNN), Logistic Regression and Random Forest (RF), to forecast customer satisfaction scores of customers using the sales dataset from the Brazilian e-commerce company, Olist. To categorise customer satisfaction scores, the results were divided into Low, Average, Good and Excellent categories. The authors use a feature engineering method to develop new features combining delivery duration and mean product rating value due to the significant impact the two factors have on the satisfaction score of purchasers. The results showed a macro F1 score of 0.32, recall of 0.36 and precision of 0.34 with Random Forest performing the best and a substantial portion of the sales orders receiving Excellent scores. This paper extends existing business analytics research by introducing feature engineering to improve model performance through combining the features with the high influence for prediction.

Moon et al. (2021) explores how several factors, including on time delivery, return policy, quality and price per market share, can have an impact on customer satisfaction and behaviour during online shopping. The paper used 40,000 data and together with Random Forest classification, Decision Tree, Naive Bayes and Apriori algorithms to conduct the research. Results show that Apriori achieved the most optimal results with an accuracy of 88% followed by Naive Bayes with an accuracy of 87%. The paper provides contribution to the field of business analytics by investigating the development of business intelligence to improve satisfaction in online shopping experience. However, the weakness of the paper lies in the limited use of advanced machine learning as it only explores traditional algorithms without comparing with new model approaches. It also had limited features as it only includes structured customer and shopping variables without social media or clickstream data.

2.2 Business Analytics, Dashboard, and Decision Support

Zimmermann and Brandtner (2024) proposed a dashboard framework that was infused with machine learning so it can be used to make business decisions in supply chain management. Descriptive analytics, RFM analytics, market basket analytics, TURF analysis, K-means clustering, and demand forecasting were combined in this study so that interactive Power BI dashboards can be

generated. The dashboard uses the transaction data from an Australian retailer, and it is mainly for executives, sales representatives, and marketing managers. Users without expertise in machine learning are still able to interpret complex analytical results from the dashboard and make better decisions related to customer segmentation, inventory management, cross-selling and marketing campaigns. From the findings, it can be understood that business intelligence can be enhanced, decision-making can be improved, and customer satisfaction can be increased by integrating outputs of the machine learning with the interactive dashboards. This study is relevant with the present research as it highlights how techniques related to business analytics and visualization of dashboards can transform transaction data into actionable managerial insights.

While Zimmermann and Brandtner (2024) focuses on how interactive dashboards can be used in practical, Alfadhel (2025) uses a longitudinal study on 1,150 publicly listed Saudi Arabian retailers from 2019 to 2023 to highlight how business analytics capability (BAC) can influence the performance of the firm and customer satisfaction in the retail industry. The study examined if business analytics helps organizations to improve their operational performance and customer satisfaction under different industry circumstances. This study was guided by the Dynamic Capabilities Theory. It is proven that when the business analytics capabilities are stronger, it can significantly improve both the firm performance and customer satisfaction. It was observed that it has a greater impact on market-based performance measures compared to accounting-based measures. In addition, business analytics has a stronger positive effect in business environments that are more dynamic and complex. This highlights how important it is to align the analytics capabilities with organizational strategies and industry conditions. By investing in business analytics, it can provide organizations with a sustainable competitive advantage. This can occur by supporting better decision-making, improving customer satisfaction and enhancing the business performance in long-term.

Focusing on advanced data integration, Zhang and Guo (2024) proposed a framework involving the integration of big data and multimodal data analysis to predict consumer behaviour and satisfaction as well as to understand the influence behind different attributes such as product information, behavioural data and geographic location. The study found that multimodal approaches perform better and more accurately in prediction compared to unimodal approaches by analysing the contrast in cross border e-commerce data processing between unimodal and multimodal methods. Additionally, the results showed that implementing big data analytics together with multimodal learning models can help in examining customer e-commerce data across different counties, serving improvements in decision-making based on live data insights. Therefore, this contributes greatly to e-commerce and business analytics as it utilises both big data analytics and multimodal methods to explore how large-scale business data and multimodal data can be used to capture different aspects of customer behaviour and customer satisfaction. Despite these benefits, several limitations to the study exist including the high computational complexity as multimodal approaches require large volumes of labelled multimodal data, increase training times and higher complexity in model architecture.

In contrast to multimodal predictive approaches, the research by Griva (2022) applies a business analytics methodology to segment customers using customer satisfaction data from 270000 survey responses across 140 e-commerce platforms. Customer satisfaction scores were evaluated at two stages which are during checkout and after product delivery. Through this framework, the author identified customer segments to provide personalised marketing initiatives including promotional and referral campaigns for dissatisfied and satisfied customers respectively. These initiatives were met with positive results after six months as customer satisfaction improved, indicating the effectiveness of customer segmentation in improving customer satisfaction. However, the study only used unsupervised learning methods and survey data from a single European country, introducing limitations to this study as it can only identify the customer segments and cannot predict future

customer satisfaction at the same time may not be able to generalise well to other regions. Despite these limitations, this research still contributes to business analytics as it uses a large-scale dataset to make the findings reliable and representative. Not only that, the paper also links customer segments to personalised marketing actions showing the real-life practical application to the results.

2.3 Customer Behaviour, Segmentation and Business Intelligence

Feng (2025) examines the role of AI and machine learning in e-commerce, the study analysed three different applications, customer purchase prediction, customer segmentation, and market basket analysis. In the customer purchase prediction analysis, several machine learning models were applied, including Decision Tree, Random Forest, Logistic Regression, and Gradient Boosted Trees. The results showed that the Decision Tree model achieved the highest accuracy of 88.8%, with annual salary and age identified as important factors influencing purchasing decisions. For the second analysis, the study applied K-means clustering to segment customers based on demographic and purchasing behaviour. The findings revealed that customers could be grouped into different segments with distinct purchasing patterns, allowing businesses to better understand customer needs and develop targeted strategies to improve customer satisfaction and encourage repeat purchases. Lastly, the study of market basket analysis uses association rules to identify relationships between products and customer purchasing patterns. The findings revealed that products such as ice cream and butter, as well as milk and chocolate, were frequently purchased together, indicating strong opportunities for cross-selling. However, the study has several limitations. The analysis was limited by the use of fewer features in purchase prediction and the lack of a clear method for determining the optimal number of customer clusters. Additionally, the evaluation of association rules relied mainly on interpretation rather than detailed performance metrics, which may affect the reliability of the findings.

While Feng (2025) covered several applications of customer analytics, Kakani et al. (2021) investigate customer retention in e-commerce by applying big data and machine learning techniques, specifically Random Forest (RF), to predict customer churn. Using a Kaggle dataset, the study employed preprocessing steps such as handling missing values, removing outliers, one-hot encoding, Z-score normalization, and Synthetic Minority Oversampling Technique (SMOTE) to address class imbalance. The RF model achieved strong results with 95% accuracy, 98% precision, 83% recall, and a ROC-AUC of 98.51%, outperforming decision trees and support vector machines. These findings highlight the effectiveness of ensemble learning in managing large, complex datasets and improving retention strategies through predictive analytics. The paper contributes to business analytics by demonstrating how predictive modeling can optimize customer lifetime value and reduce churn-related losses. While RF proved highly effective, the study acknowledges limitations such as reliance on a single dataset and the need for deeper exploration of real-time and deep learning approaches. The research gap lies in scaling predictive retention models for broader deployment and integrating advanced techniques beyond ensemble learning to further enhance customer retention in dynamic e-commerce environments.

Beyond individual prediction models, Madanchian (2024) reviews how complex systems theory can enhance predictive analytics in e-commerce, particularly in business management. The paper emphasizes that traditional linear models often fail to capture the dynamic, non-linear, and emergent behaviours of e-commerce systems, such as supply chain variability, customer behaviour, and market competition. By integrating approaches like agent-based modeling, network theory, and systems dynamics, predictive analytics can better handle interconnectedness, feedback loops, and adaptability. Applications discussed include inventory optimization, dynamic pricing, fraud detection, and personalized customer experiences, with examples from companies like Amazon, Netflix, and Walmart. The study contributes to business analytics by highlighting the need to embed predictive models within a complex systems framework to improve resilience and strategic decision-making.

While machine learning and deep learning methods (e.g., Random Forests, LSTM, ARIMA) are widely used, the review identifies a research gap in connecting these tools with broader organizational and environmental complexities. Future research is suggested in areas such as ethical data use, real-time analytics integration, and hybrid modeling to address challenges like data sparsity, noise, and scalability in e-commerce predictive systems.

2.4 Sales and Demand Forecasting

Debbagh et al. (2025) proposed a deep learning approach to improve sales demand forecasting using the Olist Brazilian e-commerce dataset, which contains about 100,000 orders from 2016 to 2018. The objective of the study was to identify the best deep learning model for predicting future sales. Before training, the data were cleaned and enhanced using feature engineering, including Recency, Frequency, Monetary (RFM) analysis, product seasonality and Brazilian public holidays. The study compared six deep learning models including LSTM, GRU, TCN, CNN-LSTM, CNN-GRU and LSTM-Attention. The results showed that LSTM-Attention achieved the best performance with the lowest MSE (0.0045) and RMSE (0.0671) because the attention mechanism helped the model focus on the most important information. This was followed by the standard LSTM model which was able to learn long-term patterns in the data but had higher prediction errors compared to LSTM-Attention. The GRU model had a simpler structure and trained faster while the TCN model was able to capture time patterns effectively using convolutional layers but both forecasting accuracy was lower than LSTM-Attention achieved. Lastly, the hybrid CNN-LSTM and CNN-GRU models combined feature extraction from CNN with sequence learning from LSTM or GRU to perform better than some individual models. The study only compared deep learning models and did not evaluate traditional machine learning methods. It also did not consider external factors such as economic conditions and the LSTM-Attention model requires high computational resources that may be less suitable for real-time applications.

In contrast, Panga (2023) explores the application of machine learning models including Linear Regression, Polynomial Regression, Random Forest, and Gradient Boosting, for forecasting e-commerce sales and demand. The study emphasizes the importance of accurate predictions in optimizing inventory management, reducing costs, and improving customer satisfaction. Using historical sales data, promotional activity, and external signals, the research applies data preprocessing, exploratory data analysis, and feature engineering to prepare datasets for predictive modeling. Performance was evaluated using metrics such as MAE, RMSE, and R^2 , with Gradient Boosting emerging as the most accurate model, followed by Random Forest, while linear and polynomial regression served as simpler baselines. The paper contributes to business analytics by comparing multiple forecasting approaches within the same e-commerce context, highlighting both their strengths and limitations. While ensemble methods provided superior accuracy, the study identifies a research gap in the limited exploration of hybrid models that combine different algorithms to leverage their complementary strengths. This comparative framework offers practical insights for businesses seeking to refine marketing strategies and inventory control through advanced analytics.

Besides that, Huo (2021) examines sales prediction in e-commerce using a comparative approach across linear models, machine learning models, and deep learning models. The study uses Walmart's M5 dataset, which includes over 3,000 products across three states, and evaluates models such as ARIMA, exponential smoothing, Random Forest, XGBoost, MLP, and LSTM. Interestingly, the findings reveal that linear models, especially exponential smoothing and ARIMA, often outperform more complex machine learning and deep learning approaches in terms of accuracy and efficiency. However, incorporating calendar and price-related features significantly improves prediction performance across all models, highlighting the importance of feature engineering in business analytics. This work contributes to e-commerce analytics by showing that while advanced algorithms

are popular, simpler statistical models can be more effective for structured retail sales data. The research gap identified is the limited exploration of additional influencing factors like customer reviews, search trends, promotions, and the scalability of models across larger datasets. Future studies are encouraged to integrate broader indicators and test whether machine learning and deep learning approaches gain advantages in more complex or larger-scale scenarios.

2.5 Delivery Performance and Customer Feedback Analytics

Rahib et al. (2024) study three important e-commerce metrics, customer churn, annual customer spending, and product on-time delivery. Customer churn can negatively result in revenue and customer loyalty, while delivery problems can negatively affect e-commerce growth. Therefore, the study aims to provide useful insights to improve customer retention, increase revenue, and optimize delivery performance. The study utilizes three different datasets obtained from Kaggle. For model development, the datasets were split into training and testing sets, with a 75:25 ratio for customer churn and annual spending, and a 70:30 ratio for product on-time delivery. For customer churn and product on-time delivery prediction, several machine learning models, including Support Vector Machine (SVM), Decision Tree, Random Forest, and XGBoost, were applied. Meanwhile, Linear Regression was used for customer annual spending prediction. The results show that SVM achieved the best performance for customer churn prediction with an accuracy of 83.45% and product on-time delivery prediction with an accuracy of 68.42%. Additionally, Linear Regression achieved an R^2 score of 0.9782 on the test data for annual spending prediction, indicating a strong model fit with low prediction errors. The study also revealed important factors influencing predictions, including purchase frequency, product categories, and average session length.

Next, Sugiyanto et al. (2025) evaluated the application of natural language processing (NLP) and machine learning techniques to analyse sentiment from e-commerce customer feedback. The study aimed to group customer feedback into positive and negative sentiments while evaluating the effect of stopword removal during the data preprocessing stage. Several text processing techniques, including tokenisation, stemming, and TF-IDF feature extraction, were applied before developing machine learning models such as Support Vector Machine (SVM) and Naïve Bayes. The results showed that SVM achieved better performance compared to Naïve Bayes, obtaining the highest accuracy, which is 99.9%, when stopword removal was excluded during preprocessing. The findings suggest that removing common words does not always improve model performance and should be carefully considered depending on the dataset and algorithm used. Overall, the study demonstrated that machine learning-based sentiment analysis can help e-commerce businesses understand customer opinions, identify customer satisfaction levels, and use review information to support better decision-making. However, a model with extremely high accuracy could potentially be overfitting as it memorises patterns from the training data but performs poorly on new data.

Similarly, Hossain et al. (2021) studies how machine learning models can be used to predict online rating products using textual product reviews from the dataset “GrammarandProductReview” from Datafiniti. Supervised machine learning methods were used to support the study including Logistic Regression, XGBoost, Random Forest Classifier together with TF-IDF Vectoriser. The findings showed that Random Forest provided the best results with 94% accuracy, 0.94 precision, 0.94 recall and 0.94 F1-scores. In terms of study limitations, given that the models focus only on text reviews, the quality of the reviews may negatively impact the predictive performance and accuracy, especially when reviews are ambiguous or do not provide sufficient details. In spite of these weaknesses, the paper supports business analytics research as it applies natural language processing to machine learning algorithms to extract useful information from unstructured textual data. This provides practical business insights as businesses can derive customer satisfaction and opinions without solely relying on numerical ratings.

Now, by focusing on customer service interactions, the study by Gao (2025) uses a linear regression model to explore the measure of importance of manual customer service on customer satisfaction using features such as channels of communication, types of customer inquiries and duration taken for customer service to respond. The linear regression model showed negative linear correlation between customer response time and satisfaction, highlighting that the slower the response time, the lower customer satisfaction would be. Moreover, random forest was used to investigate the attributes and influence of each factor. The model had ascertained that telephone communication channels had higher effectiveness compared to emails while weekend and night communication can also lead to greater satisfaction scores. The research contributes by focusing on manual customer service interactions which is an unexplored but important determinant towards customer satisfaction and retention. However, this may also be a limitation to the study as it limits generalisation as it mainly focuses on manual customer service and used data from a single platform and region.

2.6 Comparison Table

Study	Research Method	Findings	Weaknesses
Research Focus: Customer Satisfaction Prediction			
Mangla & Soni (2025)	Random Forest, HistGradientBoosting, XGBoost, LightGBM, Voting Ensemble	Voting Ensemble achieved 82.53%; delivery delay and carrier time were major predictors.	Limited to predictive analytics; lacks dashboard visualization and real-time business support.
Zagnhoul et al. (2024)	Logistic Regression, SVM, Random Forest, Gradient Boosting, MLP	Random Forest achieved 92% accuracy after oversampling; identified logistics and payment-related predictors.	Focuses only on prediction; limited business analytics and visualization.
Wangkiat & Polprasert (2023)	KNN, Logistic Regression, Random Forest	Feature engineering improved prediction; Random Forest performed best.	Severe class imbalance; no temporal validation; moderate predictive performance.
Pandey et al. (2023)	LSTM, XGBoost, SVM, Logistic Regression, Random Forest, Naïve Bayes, Neural Network	LSTM achieved highest accuracy (87.09%); identified key satisfaction drivers.	Limited descriptive analytics, visualization, and business performance analysis.
Research Focus: Business Analytics & Dashboarding			
Zimmermann & Brandtner (2024)	Power BI, RFM, K-Means, Market Basket Analysis, Demand Forecasting	Integrated machine learning into interactive dashboards for business decision-making.	Does not include customer satisfaction prediction models.

Alfadhel (2025)	Longitudinal statistical analysis	Demonstrated that stronger business analytics capability improves customer satisfaction and firm performance.	Does not implement machine learning or predictive models.
Research Focus: Customer Behaviour & Segmentation			
Feng (2025)	Decision Tree, Random Forest, Logistic Regression, Gradient Boosting, K-Means, Association Rules	Decision Tree achieved 88.8%; customer segmentation and market basket analysis improved business insights.	Limited feature selection; unclear method for selecting optimal clusters; limited evaluation of association rules.
Gogineni (2026)	RFM, K-Means, DBSCAN, Moran's I, NLP, Logistic Regression, RF, XGBoost, SVM	Combined descriptive, spatial, NLP and predictive analytics; XGBoost achieved 92%.	Focuses on analytics framework rather than interactive business dashboards.
Griva (2022)	Business Analytics Framework, Clustering	Identified customer segments and linked them to personalized marketing strategies.	Uses only clustering and survey data; cannot predict future satisfaction.
Research Focus: Sales & Demand Forecasting			
Debbagh et al. (2025)	LSTM, GRU, TCN, CNN-LSTM, CNN-GRU, LSTM-Attention	LSTM-Attention achieved best forecasting performance using engineered features.	Compared only deep learning models; high computational requirements; ignored external factors.
Panga (2023)	Linear Regression, Polynomial Regression, Random Forest, Gradient Boosting	Gradient Boosting achieved highest forecasting accuracy; improved inventory planning.	Limited exploration of hybrid forecasting models.
Huo (2021)	ARIMA, Exponential Smoothing, Random Forest, XGBoost, MLP, LSTM	Traditional statistical models outperformed complex ML models; highlighted importance of feature engineering.	Did not integrate customer behaviour, reviews or external business indicators.
Research Focus: Delivery Performance & Churn			
Rahib et al. (2024)	SVM, Decision Tree, Random Forest, XGBoost, Linear Regression	SVM performed best for churn and delivery prediction; Linear Regression accurately predicted annual spending.	Does not investigate customer satisfaction directly.

Kakani et al. (2021)	Random Forest, Decision Tree, SVM, SMOTE	Random Forest achieved 95% accuracy and strong ROC-AUC; effective for retention prediction.	Single dataset; lacks deep learning and real-time prediction.
Research Focus: Sentiment Analysis & Feedback			
Sugiyanto et al. (2025)	TF-IDF, SVM, Naïve Bayes	SVM achieved 99.9% accuracy; demonstrated impact of preprocessing on sentiment analysis.	Possible overfitting due to extremely high accuracy; lacks validation on unseen data.
Hossain et al. (2021)	TF-IDF, Logistic Regression, Random Forest, XGBoost	Random Forest achieved 94% accuracy; demonstrated value of NLP for customer reviews.	Relies solely on text reviews; performance depends on review quality.
Research Focus: General E-commerce Systems			
Madanchian (2024)	Complex Systems Theory, Agent-Based Modelling, Network Theory	Demonstrated how complex systems improve predictive analytics and business resilience.	Conceptual review; lacks empirical machine learning implementation.
Moon et al. (2021)	Decision Tree, Random Forest, Naïve Bayes, Apriori	Compared traditional ML models; Apriori achieved highest reported accuracy.	Limited to traditional algorithms; lacks advanced ML and richer customer data.

Table 1: Literature Review Comparison Table

2.7 Research Gap

From the literature review, it is found that the application of machine learning in e-commerce has been applied in many problems within the business such as customer satisfaction prediction, sales forecasting, customer segmentation, sentiment analysis, customer churn prediction, and demand forecasting. Algorithms such as Random Forest, XGBoost, Support Vector Machine (SVM), Gradient Boosting models, LSTM and ensemble learning models are reported to have strong predictive performance in many studies. However, most of the current research has concentrated on the creation of predictive models and their performance, with less attention being paid to a thorough descriptive analysis and business visualization. The use of dashboards or business intelligence methods was included in a few studies, but they were not always integrated with predictive analytics as part of a comprehensive business analytics program.

Other studies focused on particular aspects of e-commerce, such as customer satisfaction, sales forecasting, customer attrition or sentiment analysis. Nonetheless, the studies did not provide a comprehensive assessment of business performance. There has been little focus on integrating data cleaning, exploratory data analysis, summary statistics, anomaly detection, interactive Power BI dashboards, and predictive machine learning into one analytical workflow. Although multiple machine learning models were compared in these studies, only few of them within a single analytical workflow compared the models by descriptive analytics, anomaly detection, interactive Power BI dashboards and actionable business recommendations.

Thus, this study tackles these gaps by bringing together descriptive analytics and predictive analytics based on the Olist e-commerce dataset in Brazil. The following descriptive analytics processes are applied to the data to identify business patterns and trends. The processes include data cleaning, summary statistics, anomaly detection, and interactive Power BI dashboards. The data is then used for predictive analytics, with the help of machine learning models, to generate data-driven predictions that can guide business decisions. Logistics performance, delivery management, and data-driven decision-making in the e-commerce sector can be significantly improved by obtaining valuable insights through holistic business analytics framework by integrating visual analytics with predictive modelling.

3.0 Problem Statement

Since the growth of e-commerce has increased customer expectations for fast and reliable delivery, managing deliveries efficiently has become important for business success. However, online retailers often face problems such as late deliveries, high shipping costs, inaccurate delivery estimates and logistics issues. These problems can reduce customer satisfaction, increase business costs and affect overall business performance. Therefore, analysing the factors that affect delivery performance can help businesses to improve logistics planning and make better decisions.

The Olist e-commerce platform contains a large amount of data about orders, products, customers, sellers and deliveries. This data can be analysed to find delivery patterns and identify possible problems. Without proper analysis, businesses may miss important information such as the causes of late deliveries, product categories or states with frequent delays, areas with delivery problems and orders with unusually high shipping costs. By identifying these patterns, businesses can improve delivery processes, supply chain processes, manage logistics better and provide a better customer experience in terms of delivery.

Therefore, this study focuses on analysing the Brazilian E-Commerce Public Dataset by Olist store to understand order performance and delivery efficiency. Descriptive analytics will be conducted using summary statistics and data visualisation to analyse delivery performance, including late deliveries, delivery days and freight cost. Anomaly analysis will be performed to identify unusual patterns such as extreme late deliveries, extreme early deliveries, high freight cost-to-product value ratios and unusual sales trends. Predictive analytics will also be applied to develop machine learning models, including Random Forest, Logistic Regression and XGBoost to predict whether an order will be delivered late or on time based on historical order, product and logistics information. The study also evaluates and compares the performance of different machine learning models using evaluation metrics including Accuracy, Balanced Accuracy, Recall, F1-score, ROC-AUC and PR-AUC.

4.0 Methodology

4.1 Methodology Overview

The methodology began by performing data preparation. In this phase, the selected datasets were cleaned, merged and transformed into a single analytical dataset. Data cleaning was also conducted where the missing values, duplicate records, and inconsistent data formats were handled. This was then followed by feature engineering to generate additional columns to support both descriptive and predictive analysis. Descriptive analytics were performed by using Power BI where summary statistics, and data visualizations were developed to examine the delivery performance, freight costs, customer locations, and product categories. By analyzing this, an overall understanding of the dataset was gained, and it helped to identify important business patterns. Finally, predictive analytics were performed by using Python in Google Colab where three classification models, including Logistic Regression, Random Forest, and XGBoost, were developed. These models were considered to predict whether an order will be delivered on time or late. Multiple performance metrics

such as Accuracy, Balanced Accuracy, Recall, F1-Score, ROC-AUC, and PR-AUC, were compared to determine the best model to predict late deliveries.

4.2 Data Preparation

4.2.1 Data Cleaning and Preparation

Python was selected to perform the data preparation process to ensure the dataset was suitable for descriptive analytics and predictive modelling. Five datasets, which are Orders, Order Items, Products, Customers and Product Category Translation, from the Olist E-Commerce dataset were chosen and imported using Pandas library in Google Colab. Missing values, duplicate records, and inconsistent datatypes were identified through the initial data quality assessment.

Each dataset was cleaned individually. Duplicate records in all these datasets were removed first. Then, in the Orders dataset, date-related columns were converted to datetime format. Then, by merging the Products dataset and Product Category Translation dataset, product categories that were initially in Portuguese were translated to English. Categories with no translation were retained using the Portuguese name whereas records with missing category names were labelled as 'unknown'.

Since delivery delays can only be calculated for deliveries that are completed, only orders with the Delivered status and complete delivery dates were retained. Then, the cleaned datasets were merged into one single dataset. After merging, records that contain missing categorical values were replaced with 'unknown' whereas missing numerical values were imputed using median. The final cleaned dataset contained no missing values or duplicate orders after confirming through a final quality check.

4.2.2 Feature Engineering

Descriptive analytics and predictive modelling were supported by several new variables that were generated. The primary target variable, `late_delivery`, was created by calculating `delivery_delay_days`. Meanwhile, `delivery_status`, `days_late`, and `days_early` were the added descriptive variables to support the visualization of the dashboard. By using the order purchase timestamp and estimated delivery date, temporal features such as `purchase_year`, `purchase_month`, `purchase_weekday`, `purchase_hour`, `purchase_quarter`, and `estimated_delivery_days` were extracted.

Additional variables such as `product_volume_cm3` and `item_total_value` were created at the item level. Order level features were then aggregated by these records. The features include `item_count`, `unique_product_count`, `unique_seller_count`, `unique_category_count`, `total_product_value`, `total_freight_value`, `total_product_weight_g`, `total_product_volume_cm3`, `total_order_value`, and `freight_to_price_ratio`. Dominant product category for each order were identified and stored under the `main_product_category` column. These engineered variables formed the foundation to conduct both descriptive analytics and predictive modelling.

4.3 Descriptive Analytics

4.3.1 Summary Statistics

Before summary statistics were generated for the selected numerical variables, the cleaned dataset was imported into Microsoft Power BI. Summary statistics were calculated in order to give an overview of an initial assessment of the dataset before further analysis was conducted. The statistics that were calculated include Mean, Median, Standard Deviation, Minimum and Maximum for each selected variable.

The variables analyses included Delivery Delay, Estimated Delivery Days, Freight Value and Freight-to-Price Ratio. The summary statistics for delivery delays can help provide insight into the

timeliness of shipping orders while estimated delivery days can explain the expected delivery duration. Moreover, freight value indicates the shipping cost for individual orders while total freight value estimates the total shipping costs across the dataset. Finally, the summary statistics of freight-to-price ratio was also selected to examine the relationship between shipping costs and product prices. These statistics were included to provide an overview of the selected variables by summarizing their overall distribution, central tendency and variation before further analyses and visualizations were carried out.

4.3.2 Data Visualization

To create interactive dashboards and charts for data visualization, Microsoft Power BI was also used due to its functional capability in creating interactive visualizations that allow users to explore the dataset efficiently. Visualizations can help to support exploratory data analysis by finding patterns, trends and relationships within the dataset that help answer business questions. Various visualization types were included based on variable characteristics to allow comparison across categories, distribution analysis and relational trend exploration to be conducted.

Several charts were developed such as the delivery status distribution, late deliveries by customer state, late deliveries by product category. These visualizations depict key business metrics clearly across various geographic locations and product categories, providing support in exploring the data and identifying areas that should be further explored to provide important business insights.

4.4 Predictive Analytics

4.4.1 Target and Predictor Selection

The objective of the predictive analytics component was to predict whether an order will be delivered on time or later. Hence, the variable `late_delivery` was selected as the target. Variables such as customer location, temporal features, estimated delivery period, product characteristics, freight cost, order value, and other engineered variables were chosen as the predictors. To prevent data leakage, variables such `delivery_delay_days`, `delivery_status`, `days_late`, and `days_early` that directly revealed the delivery outcome were excluded.

4.4.2 Data Preprocessing for Machine Learning

The dataset was prepared for machine learning before the model development. median was imputed for missing numerical values and 'unknown' label was assigned to missing categorical variables. One-Hot Encoding was used to transform categorical features whereas when required, numerical variables were standardized. The dataset was then divided into training and testing sets. Stratified sampling and fixed random state were used to divide the data into 80% of training data and 20% of testing data.

4.4.3 Logistic Regression

One of the models chosen was Logistic Regression as the baseline classification model due to its simplicity and interpretability. Based on the selected predictor variables, the model is able to estimate the probability of an order being delivered late. This model can provide a useful benchmark as well when comparing its performance with other machine learning models that are more advanced.

4.4.4 Random Forest

An ensemble model, which is Random Forest, was implemented as well as it consists of multiple decision trees. By randomly selecting the subsets of the observations and predictor variables, each tree was trained. However, through majority voting, the final prediction was determined. This

was to improve the stability of the prediction and reduce the risk of overfitting compared to a single decision tree.

4.4.5 XGBoost

The final model was XGBoost due to its predictive performance that is strong and its ability to model complex relationships within datasets that are structured. XGBoost builds the trees sequentially unlike Random Forest. Each new tree focuses on correcting the errors of the prediction of the previous trees. By implementing this gradient boosting approach, higher prediction accuracy can be produced and overall classification performance will be increased.

4.4.6 Model Evaluation and Selection Criteria

Evaluation metrics that were chosen to evaluate the models were Accuracy, Balanced Accuracy, Precision, Recall, F1-score, ROC-AUC, and PR-AUC. This was to provide a comprehensive assessment of classification performance, especially for datasets that are imbalanced. The model that had the best performance overall across these metrics was selected as the best and final predictive model for this study.

5.0 Anomaly Analysis

5.1 Extreme Late Delivery Anomaly

	Metric	Value
0	Maximum Delivery Delay	188 days
1	Upper Limit	28.0 days
	Delivery Delay Anomaly	Number of Orders
0	Normal Delay	6088
1	Abnormal Delay	446

Normal Delay (≤ 28.0 days)
Abnormal Delay (> 28.0 days)

	order_id	delivery_delay_days	estimated_delivery_days	main_product_category	customer_state
53963	1b3190b2dfa9d789e1f14c05b647a14a	188	20	cool_stuff	RJ
19029	ca07593549f1816d26a572e06dc1eab6	181	29	auto	ES
11074	47b40429ed8cce3aee9199792275433f	175	16	home_construction	SP
78961	2fe324feb907e3ea3f2aa9650869fa5	167	23	furniture_decor	SP
86459	285ab9426d6982034523a855f55a885e	166	29	musical_instruments	SE

Figure 1: Extreme Late Delivery Anomaly

Figure 1 shows the extremely late delivery anomalies that were identified using the `delivery_delay_days` variable. Only orders with positive values were analyzed because positive values represent deliveries that arrived after the estimated delivery date. The Interquartile Range (IQR) method was used to calculate the threshold for abnormal delays. Based on the outputs, orders with delivery delays of more than 28 days were assumed as abnormal late deliveries while delivery delays less than or equal to 28 days were assumed as normal late deliveries. There were 6088 orders considered as normal delay, 446 orders considered as abnormal delay and the first three product categories that had the highest late delivery days were `cool_stuff`, `auto` and `home_construction`.

The abnormal late deliveries represent orders that experienced longer delays than normal delayed orders. These extreme delays may show problems in the supply chain process such as seller processing delays, inventory shortages, courier issues, long-distance transportation challenges or

inaccurate delivery time estimation. Although only 446 orders were classified as abnormal delays, these delays may have a strong impact on customer satisfaction because customers experienced delays of more than 28 days beyond the expected delivery date. Identifying these late deliveries helps businesses to understand where delivery processes require improvement.

To improve delivery performance, Olist should monitor orders that are likely to be delayed and take action before they exceed the expected delivery date. The company can also check seller and courier performance regularly to identify delivery problems and improve service quality. Besides, Olist can use past delivery data to provide more accurate estimated delivery dates based on factors such as product type, seller location and customer state. Better inventory management can also help to ensure products are available, especially in peak periods, and prevent delays caused by stock shortages. For example, if its historical data shows that certain sellers or product categories often experience late deliveries or have abnormal delays, Olist can provide early alerts, recommend faster courier options or adjust delivery estimates to better reflect the expected delivery time. These improvements can reduce delivery delays, improve customer satisfaction and make delivery processes more efficient.

5.2 Extreme Early Delivery Anomaly

	Metric	Value
0	Maximum Early Delivery	147 days early
1	Lower Limit	33.0 days early
	Early Delivery Anomaly	Number of Orders
0	Normal Early Delivery	87134
1	Extreme Early Delivery	1510

Normal Early Delivery (within 33.0 days early)
 Extreme Early Delivery (more than 33.0 days early)

	order_id	delivery_delay_days	estimated_delivery_days	main_product_category	customer_state
38906	0607f0efea4b566f1eb8f7d3c2397320	-147	150	furniture_decor	SP
15341	c72727d29cde4cf870d569bf65edabfd	-140	147	market_place	MA
55454	eec7f369423b033e549c02f3c5381205	-135	156	furniture_decor	RS
83851	c2bb89b5c1dd978d507284be78a04cb2	-124	141	housewares	SP
65471	40dc2ba6f322a17626aac6244332828c	-109	117	art	RJ

Figure 2: Extreme Early Delivery Anomaly

Figure 2 shows the extreme early delivery anomalies that were identified using the `delivery_delay_days` variable. Only orders with negative values were analyzed because negative values represent deliveries that arrived before the estimated delivery date. The Interquartile Range (IQR) method was used to calculate the threshold for extreme early delivery. Based on the outputs, orders with delivery delays more than 33 days early were assumed as extreme early delivery while early delivery within 33 days were assumed as normal early delivery. There were 87134 orders considered as normal early delivery, and 1510 orders considered as extreme early delivery. The top 3 product categories that had the highest early delivery days were `furniture_decor`, `market_place` and `housewares`.

Extreme early delivery anomalies represent orders that were delivered much earlier than the estimated delivery date. While early delivery provides a positive customer experience, these unusually early deliveries may show that the estimated delivery dates are not accurate. Inaccurate delivery estimates can make it difficult for the business to plan delivery schedules, measure courier performance

and provide reliable delivery information to customers. Over time, this may reduce customer confidence in the estimated delivery dates provided by the platform.

To improve delivery estimation accuracy, Olist should regularly review how estimated delivery dates are calculated and use historical delivery data to create more accurate delivery predictions. The company can analyze previous delivery patterns such as average delivery time by seller location, product category and customer region to provide more realistic, estimated dates. Olist can also monitor unusually early deliveries to identify possible data recording errors or inaccurate delivery estimates. For example, if a product category is consistently delivered 20 days earlier than the estimated date, Olist can adjust the estimated delivery time for that category to better match the actual delivery date. These improvements can help customers to receive more reliable delivery information, improve trust in the platform and support better delivery planning.

5.3 High Freight Cost Ratio Anomaly

Metric		Value
0	Maximum Freight-to-Price Ratio	21.45
1	Upper Limit (IQR)	0.75
Freight Cost Anomaly		Number of Orders
0	Normal Freight Ratio	89749
1	High Freight Ratio Anomaly	6385

Normal Freight Ratio (≤ 0.75)						
High Freight Ratio Anomaly (> 0.75)						
order_id	total_product_value	total_freight_value	freight_to_price_ratio	main_product_category	customer_state	
80790	6e864b3f0ec71031117ad4cf46b7f2a1	0.85	18.23	21.447059	construction_tools_construction	RJ
68968	3ee6513ae7ea23bdfab5b9ab60bffc5	0.85	18.23	21.447059	construction_tools_construction	SP
81879	7c92284adbec8033d160303d83065dbf	9.18	54.69	5.957516	housewares	RS
14709	b2d30dfaf7114de8ab3f737d062032fe	5.95	34.15	5.739496	signaling_and_security	BA
71965	21fb84496c034392ca2aaf7f7919a3f	6.99	37.90	5.422031	housewares	PE

Figure 3: High Freight Cost Ratio Anomaly

Figure 3 shows the high freight cost ratio anomalies that were identified using the `freight_to_price_ratio` variable. The Interquartile Range (IQR) method was used to calculate the threshold for detecting unusually high freight costs compared with the product value. Based on the outputs, orders with a freight-to-price ratio greater than 0.75 were assumed as high freight cost anomalies, while freight-to-price ratio equal or lower than 0.75 were assumed as normal freight ratio. There were 89749 orders considered as normal freight ratio while 6385 orders considered as high freight cost anomalies. The top 3 product categories that had the highest freight cost ratio were `construction_tools_construction`, `housewares` and `signaling_and_security`.

High freight-to-price ratio anomalies show that some orders have shipping costs that are too high compared with the product value. This can affect customer satisfaction, especially when customers buy low-priced products but need to pay high delivery fees. These situations may be caused by long delivery distances, large product sizes, or inefficient shipping methods.

To reduce high freight cost issues, Olist can improve delivery route planning, choose suitable courier services based on product and location and review shipping fees for low-price products to make adjustments if required. The company can also analyze high freight cost by region and product category to identify ways to reduce delivery costs. For example, if certain low-priced products in a specific region have high freight-to-product value ratios due to expensive delivery costs, businesses can consider using alternative courier services, improving delivery routes or offering more suitable

shipping options to reduce unnecessary costs. This can help to reduce shipping costs, maintain efficient delivery services and improve customer satisfaction.

5.4 Average Late Delivery Days by Product Category

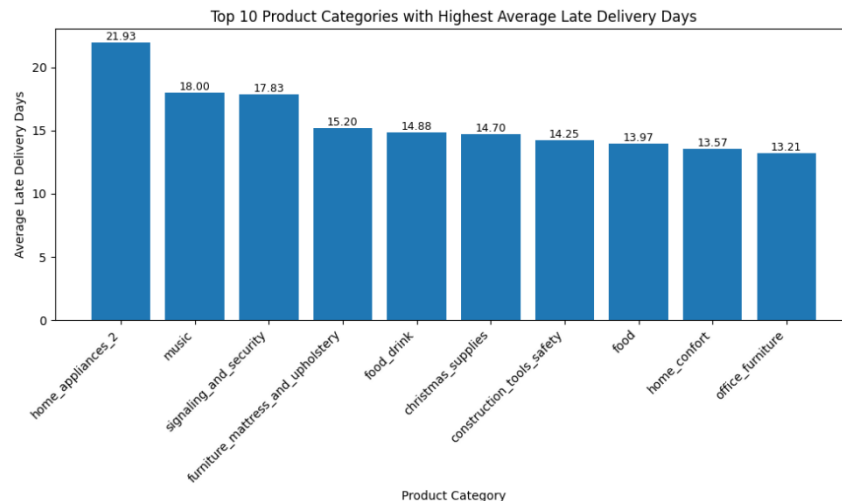


Figure 4: Top 10 Product Categories with the Highest Average Late Delivery Days

Figure 4 shows bar chart of the top 10 product categories with the highest average late delivery days. The top three categories were home_appliances_2, with an average delay of 21.93 days, followed by music with 18 days and signaling_and_security with 17.83 days. This anomaly is important because these categories experience longer delivery delays compared with other products. This may be due to issues in the supply chain, such as product availability, seller processing time, product size or weight, or transportation issues.

The results show that certain product categories have a higher risk of delayed delivery. For example, home appliances may require more careful handling, larger storage space or special transportation arrangements, which can increase delivery time. Categories with long average delays may reduce customer satisfaction because customers may need to wait much longer than expected. Identifying these categories helps Olist to understand which products require more attention and improve delivery performance.

To reduce delivery delays for these product categories, Olist should monitor the performance of sellers and courier services involved with high-delay products. The company can improve inventory planning, work with more suitable delivery partners and adjust estimated delivery times based on product characteristics. For example, if furniture products often experience longer delivery days due to large size or transportation difficulties, Olist can assign specialized courier services for these products, maintain sufficient stock levels and provide customers with more realistic delivery estimates. These actions can provide more accurate delivery expectations, reduce delay risks and improve customer experience and satisfaction.

5.5 Monthly Order Value Trend (2018)

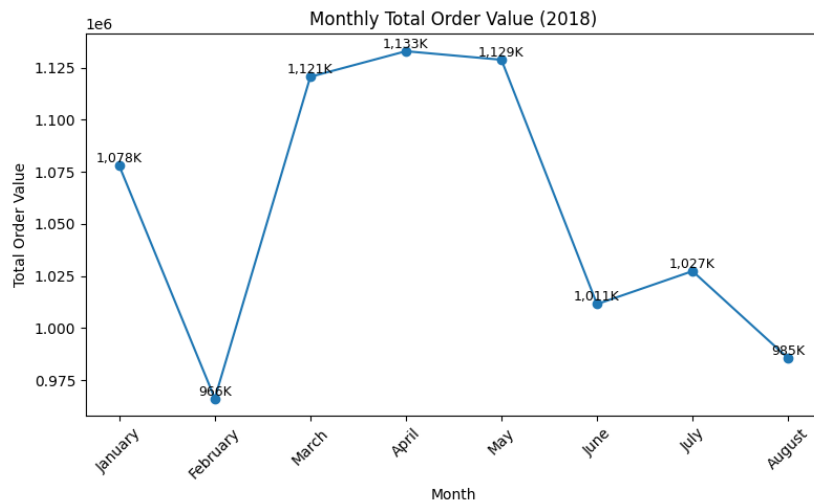


Figure 5: Monthly Order Value Trend (2018)

Figure 5 shows line chart of the monthly total order value trend in 2018. From January to February, the total order value decreased from 1,078K to 966K, showing a reduction in customer spending and sales performance. This decrease may be caused by changes in customer purchasing behavior, lower demand during that period or fewer high-priced product purchases. From February to March, the total order value increased from 966K to 1,121K, which shows an improvement in sales performance. This increase may be caused by higher customer demand, more promotions or events or the number of orders increases. From May to June, the total order value decreased again from 1,129K to 1,011K. This drop may indicate a temporary decrease in customer purchases or changes in market demand.

The changes in monthly order value show that customer purchasing behavior is not constant and can change across different periods. These unusual increases and decreases in sales trends are important to analyze because they help businesses to understand changes in customer demand and identify periods with weaker sales performance. Sudden drops in order value may affect revenue planning, inventory management and marketing strategies if they are not monitored.

To manage sales fluctuations, Olist should monitor sales trends regularly and identify factors that cause sudden changes in order value. The company can use historical sales data to improve demand forecasting, plan inventory levels more effectively and develop targeted promotions during periods with lower sales. For example, if the analysis shows that sales usually decrease in certain months, Olist can offer promotions or discounts to encourage more customers to purchase. The company can also prepare sufficient stock for months with higher demand to avoid stock shortage. Understanding customer purchasing patterns can help Olist to maintain high sales performance and make better business decisions.

6.0 Results and Discussion

6.1 Descriptive Analytics Results

6.1.1 Summary Statistic Results

Summary Statistics of Delivery Delay	
Statistic	Value

Mean	-11.88
Median	-12
Standard Deviation	10.18
Minimum	-147
Maximum	188

Table 2: Summary Statistics of Delivery Delay

Table 2 presents the summary statistics for delivery delay. A negative delivery delay indicates that an order was delivered before its estimated delivery date, while a positive delivery delay indicates that it was delivered after the estimated date. The mean delivery delay was -11.88 days, with a median of -12 days, suggesting that most orders were delivered approximately 12 days earlier than expected. The standard deviation of 10.18 days indicates moderate variation in delivery performance. Delivery delays ranged from -147 days (147 days earlier than the estimated delivery date) to 188 days (188 days later than the estimated delivery date), indicating the presence of both exceptionally early and significantly delayed deliveries.

Summary Statistics of Estimated Delivery Days	
Statistic	Value
Mean	24.37
Median	24
Standard Deviation	8.76
Minimum	3
Maximum	156

Table 3: Summary Statistics of Estimated Delivery Days

Table 3 shows the summary statistics for estimated delivery days, which represent the number of days between the purchase date and the estimated delivery date provided to customers. The mean estimated delivery period was 24.37 days, with a median of 24 days, indicating that customers were generally given an estimated delivery time of around 24 days. The standard deviation of 8.76 days indicates moderate variability in the estimated delivery periods across orders. The estimated delivery time ranged from 3 to 156 days, reflecting differences in expected delivery durations based on factors such as product type, seller location, or shipping conditions.

Summary Statistics of Freight Value	
Statistic	Value
Mean	22.79
Median	17.17
Standard Deviation	21.56

Minimum	0
Maximum	1,790

Table 4: Summary Statistics of Freight Value

Table 4 presents the summary statistics for total freight value, which represents the combined shipping cost of all items within the same order. The average total freight value was 22.79, while the median was 17.17, indicating that most orders had relatively lower shipping costs, while a smaller number of orders incurred substantially higher shipping charges, resulting in a positively skewed distribution. The standard deviation of 21.56 indicates considerable variation in shipping costs across different orders. The total freight value ranged from 0 to 1,790, suggesting the presence of orders with exceptionally high shipping costs.

Summary Statistics of Freight-to-Price Ratio	
Statistic	Value
Mean	0.31
Median	0.22
Standard Deviation	0.31
Minimum	0
Maximum	21.45

Table 5: Summary Statistics of Freight-to-Price Ratio

Table 5 presents the summary statistics for the freight-to-price ratio, which measures the proportion of total shipping cost relative to the total product value of an order. The ratio was calculated by dividing the total freight value by the total product value. The mean ratio was 0.31, while the median was 0.22, indicating that shipping costs accounted for approximately 31% of the product value on average. The difference between the mean and median suggests a positively skewed distribution, where some orders had unusually high shipping costs compared to their product value. The standard deviation of 0.31 indicates considerable variation in shipping cost proportions across orders. The ratio ranged from 0 to 21.45, suggesting that some low-value orders had shipping costs that were substantially higher than their product value.

6.1.2 Visualization

6.1.2.1 Delivery Status Distribution

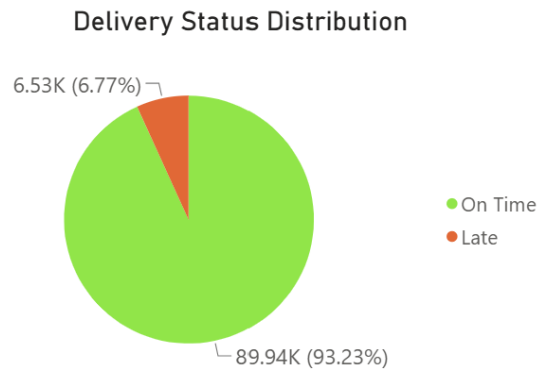


Figure 6: Delivery Status Distribution

Figure 6 presents the distribution of delivery status among all orders in the dataset. The delivery status is categorized into two classes, On Time and Late, which represent whether an order was delivered on or before the estimated delivery date or exceeded the estimated delivery date, respectively. The delivery status distribution shows that 89.94 thousand orders (93.23%) were delivered on time, while 6.53 thousand orders (6.77%) were classified as late deliveries. This indicates that the majority of orders were successfully delivered within the expected timeframe, demonstrating overall reliable delivery performance. However, a small proportion of orders experienced delays, which may be caused by factors such as longer shipping distances, logistical constraints, product availability issues, or differences in seller and courier performance. These delayed orders provide valuable insights for further analysis to identify factors that may contribute to late deliveries and improve delivery prediction.

6.1.2.2 Late Deliveries by Customer State

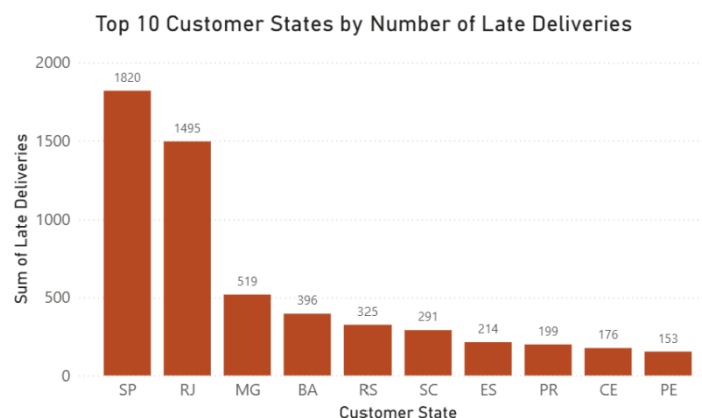


Figure 7: Top 10 Customer States by Number of Late Deliveries

Figure 7 presents the top 10 customer states with the highest number of late deliveries. The number of late deliveries was calculated by summing the binary late_delivery variable, where a value of 1 represents an order delivered after the estimated delivery date. Among the selected states, SP recorded the highest number of late deliveries with 1,820 orders, followed by RJ with 1,495 orders and MG with 519 orders. The higher number of late deliveries observed in these states may be related to their larger customer markets and higher order volumes, which result in a greater number of shipments requiring processing and transportation. In addition, states with high shipment activity may experience greater pressure on logistics operations, including order handling, transportation capacity, and courier workload, which could contribute to delivery delays. These findings suggest that customer location and regional delivery conditions may be relevant factors when predicting late deliveries.

6.1.2.3 Late Deliveries by Product Categories

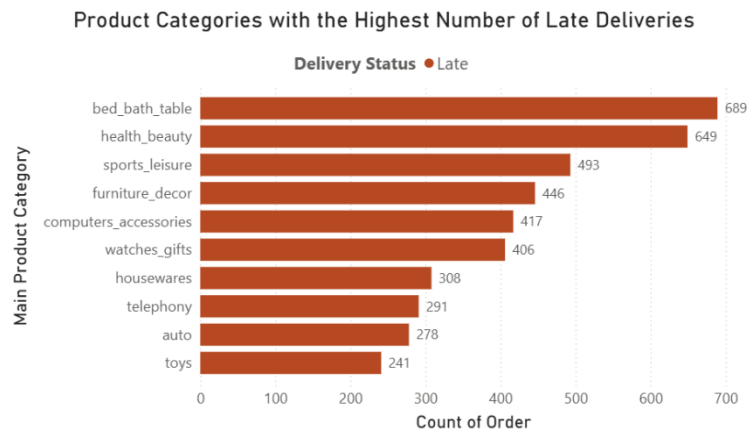


Figure 8: Product Categories with the Highest Number of Late Deliveries

Figure 8 presents the top 10 product categories with the highest number of late deliveries. The analysis is done by filtering orders with a late delivery status and counting the number of delayed orders for each product category. Among the selected categories, Bed & Bath recorded the highest number of late deliveries with 689 orders, followed by Health & Beauty with 649 orders and Sports & Leisure with 493 orders. The high number of late deliveries in the Bed & Bath category may be associated with larger and heavier items, such as furniture and household products, which may require more complex transportation arrangements. For Health & Beauty products, delays may occur due to product availability issues, stock replenishment, or additional processing time before shipment. Meanwhile, Sports & Leisure products may experience delays due to variations in product sizes, availability, and supplier fulfilment processes. These findings suggest that product characteristics may influence delivery performance and should be considered when predicting late deliveries.

6.2 Predictive Analytics Results

6.2.1 Model Performance Comparison

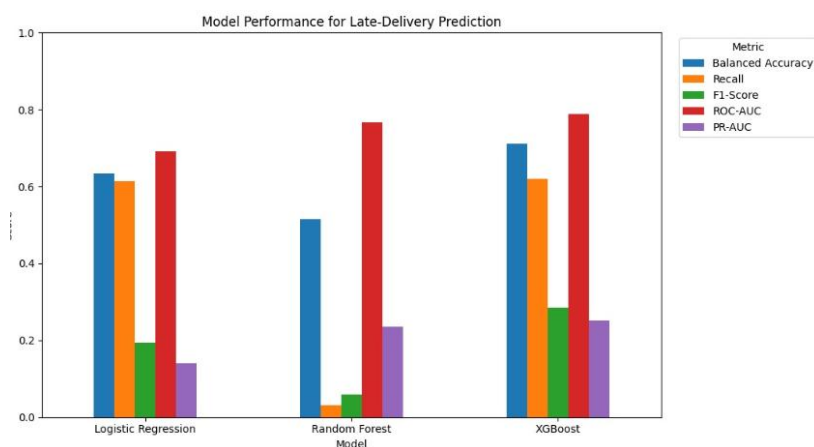


Figure 9: Model Performance Comparison Bar Chart

Model	Accuracy	Balanced Accuracy	Recall	F1-score	ROC-AUC	PR-AUC
-------	----------	-------------------	--------	----------	---------	--------

Logistic Regression	65.31%	63.48%	61.36%	0.1933	0.6914	0.1399
Random Forest	93.24%	51.46%	3.14%	0.0592	0.7666	0.2346
XGBoost	78.92%	71.06%	61.97%	0.2849	0.7894	0.2520

Table 6: Model Performance Comparison Table

Table 6 presents the performance comparison of the three classification models including Logistic Regression, Random Forest, and XGBoost, while Figure 9 provides a visual representation of the same results to facilitate comparison. The evaluation is based on six performance metrics: accuracy, balanced accuracy, recall, F1-score, ROC-AUC, and PR-AUC. Each metric provides a different perspective on the model's predictive performance, allowing for a more comprehensive comparison.

Accuracy represents the overall proportion of correctly classified instances among all observations. Although it is one of the most commonly used evaluation metrics, it may provide an overly optimistic result when the dataset is imbalanced, as a model can achieve high accuracy by correctly predicting only the majority class.

Balanced accuracy addresses this limitation by giving equal importance to both classes, making it a more reliable measure for imbalanced datasets. Recall measures the model's ability to correctly identify positive cases, indicating the proportion of actual positive instances that are successfully detected. This metric is particularly important when the objective is to minimize missed positive cases.

The F1-score provides a balanced assessment by considering both precision and recall, making it useful when both false positives and false negatives are important. Meanwhile, the ROC-AUC evaluates the model's capability to distinguish between the two classes across different classification thresholds, with higher values indicating better discrimination performance. Lastly, the PR-AUC summarizes the trade-off between precision and recall and is especially suitable for imbalanced classification problems, where it provides a more informative evaluation than accuracy alone.

6.2.2 Model Evaluation

Based on the evaluation results, XGBoost demonstrated the best overall performance among the three classification models. Although its accuracy of 78.92% was lower than that of Random Forest (93.24%), it provided a much better balance between correctly classifying both the majority and minority classes, as reflected by its highest balanced accuracy of 71.06%. This indicates that XGBoost was less biased towards the majority class and was more capable of handling the imbalanced nature of the informative dataset.

In terms of identifying positive cases, XGBoost achieved the highest recall of 61.97%, which was slightly higher than Logistic Regression (61.36%) and substantially better than Random Forest (3.14%). The extremely low recall of Random Forest suggests that, despite its impressive overall accuracy, the model failed to detect most of the positive cases. This indicates that Random Forest relied heavily on predicting the majority class, making it less suitable for this classification task.

XGBoost also recorded the highest F1-score (0.2849), outperforming Logistic Regression (0.1933) and Random Forest (0.0592). This shows that it achieved a better balance between precision and recall, resulting in more reliable predictions overall. Although Logistic Regression demonstrated a reasonable ability to identify positive cases, its lower accuracy, balanced accuracy, and F1-score indicate that its overall predictive performance was still weaker than that of XGBoost.

Similar trends can be observed in the ROC-AUC and PR-AUC results. XGBoost achieved the highest ROC-AUC (0.7894), indicating that it had the strongest ability to distinguish between the two classes across different classification thresholds. It also obtained the highest PR-AUC (0.2520), suggesting that it maintained a better trade-off between precision and recall when dealing with the imbalanced dataset. In comparison, Logistic Regression recorded the lowest ROC-AUC (0.6914) and PR-AUC (0.1399), while Random Forest achieved slightly better values (0.7666 and 0.2346, respectively) but was still limited by its inability to identify positive cases effectively.

Overall, while Random Forest achieved the highest accuracy, its extremely low recall indicates that it was ineffective in detecting late deliveries, which is the primary objective of this study. Logistic Regression provided more balanced results than Random Forest but was consistently outperformed by XGBoost across most evaluation metrics. Therefore, XGBoost was selected as the final model because it offered the most balanced and reliable predictive performance for this classification task.

6.3 Business Implications

The descriptive analysis helps Olist understand its overall delivery performance and identify areas that need to be improved. Summary statistics provide useful information about delivery days, freight costs and freight-to-price ratios, while the visualizations show delivery performance across different customer states and product categories. By understanding these patterns, Olist can identify which regions, and product categories had more late deliveries. So that Olist can monitor shipping costs, improve delivery planning and allocate logistics resources more effectively. These insights can help the company improve delivery efficiency, reduce costs and provide better customer experience.

The prediction model can help Olist to identify orders that are likely to be delivered late before the delivery happens. This allows the company to take action early instead of waiting for customers to report delivery problems. Without a prediction model, Olist would only know that an order is delayed after the estimated delivery date has passed, which may affect customer satisfaction. With the prediction model, Olist can analyze information such as seller location, product category, delivery time, freight cost and past delivery patterns to identify orders that have a high risk of being delayed. By predicting late deliveries earlier, Olist can improve delivery performance and reduce the number of delayed orders. Therefore, when a new order comes in, the model can use the information available at that time to predict whether the order is likely to be delivered late or on time.

After that, managers can use the prediction results to monitor high-risk orders and give them higher priority during packing and shipping. Customers can also be informed early if a delay is expected. This helps them to have more realistic delivery expectations and improve their shopping experience. Besides, Olist can adjust estimated delivery dates based on customer state and product category to provide more accurate delivery information. The findings can also help Olist to identify sellers and courier services that frequently have delivery delays. Managers can review their performance and work with them to improve delivery efficiency. Orders with unusually high shipping costs compared with the product value can also be flagged so that Olist can review delivery routes, choose more suitable courier services or adjust shipping charges if required.

Overall, these findings are important because they help Olist to improve delivery planning, reduce delivery delays, reduce logistics costs and improve customer satisfaction. By using the prediction model and the insights from the analysis, managers can make better business decisions and improve the delivery process.

7.0 Conclusion

This study analyzed the Brazilian Olist E-Commerce dataset to evaluate delivery performance and identify factors that may affect logistics efficiency. With the continuous growth of e-commerce, businesses face challenges in managing large volumes of orders while maintaining reliable delivery services. Therefore, this study applied business analytics and machine learning techniques to transform e-commerce data into meaningful insights that can support better operational decision-making.

The study integrated descriptive analytics, anomaly analysis, and predictive analytics to investigate delivery patterns and identify potential business issues. The results showed that the majority of orders were delivered on time, with 93.23% of orders classified as on-time deliveries. However, late deliveries remain an important concern as they can affect customer satisfaction and business reputation. The anomaly analysis identified several operational issues, including extremely late deliveries, unusually early deliveries, high freight-to-price ratios, and fluctuations in monthly order value. To address these issues, recommendations such as improving delivery time estimation using historical data, monitoring seller and courier performance, optimizing shipping routes and costs, strengthening inventory planning, and implementing targeted promotions during low-sales periods were proposed. These recommendations can help businesses to improve logistics efficiency and enhance customer experience.

For predictive analytics, Logistic Regression, Random Forest, and XGBoost were developed to predict whether an order would experience a late delivery. Although Random Forest achieved the highest accuracy, it had difficulty identifying delayed orders due to its low recall performance. XGBoost achieved the strongest overall performance by obtaining the highest balanced accuracy, recall, F1-score, ROC-AU and PR-AUC, making it the most suitable model for this study. Overall, this study demonstrates that combining descriptive analytics, anomaly detection, visualization, and machine learning can help e-commerce businesses identify operational challenges, improve logistics planning, and make more informed data-driven decisions.

7.1 Future Improvements

For future improvements, the predictive model can be further enhanced by applying techniques such as hyperparameter tuning, feature selection, and methods to handle imbalanced data. Since late deliveries represent a smaller proportion of the dataset, future studies can explore techniques that give more importance to identifying delayed orders, such as adjusting the model training process or balancing the dataset. These approaches can improve the model's ability to detect potential delivery problems and provide businesses with more reliable early warnings.

Future studies can also include additional factors that may influence delivery performance. The current study mainly focused on information available in the Olist dataset, including order details, product information, customer information, and freight costs. However, external factors such as weather conditions, traffic conditions, public holidays, and courier performance may also contribute to delivery delays. Including these variables can provide deeper insights into the causes of delivery issues and improve prediction accuracy.

Lastly, the scope of this study can be expanded by incorporating other areas of e-commerce analytics, such as customer satisfaction prediction, demand forecasting, customer segmentation, and sentiment analysis using customer reviews. The Power BI dashboard can also be improved by integrating real-time data updates and automated alerts. These improvements can allow businesses to monitor performance continuously and respond to operational challenges more effectively.

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List of Members' Contributions

Members' Name	Contribution
Keertana A/P Subramaniam (23109614)	• Conducted Literature Review. • Identified and developed the Research Gap. • Performed data cleaning and preparation of Olist datasets. • Developed and implemented the machine learning models. • Prepared the Methodology for the Machine Learning, including preprocessing, model development, and evaluation criteria.
Ayu Wen Li (22017867)	• Prepared the Abstract by summarizing the study, methodology, and key findings. • Conducted Literature Review. • Developed the Comparison Table for Literature Review. • Prepared the methodology for descriptive analytics using Power BI. • Documented the summary statistics and data visualization methods used in Power BI.
Soong Shuki (23105059)	• Prepared the Introduction and project background. • Conducted Literature Review. • Prepared the machine learning model performance comparison. • Evaluated and interpreted the performance of Logistic Regression, Random Forest, and XGBoost. • Prepared the discussion and justification for selecting the best-performing machine learning model.
Siow Qi Yung (22053037)	• Conducted Literature Review. • Developed the Problem Statement. • Conducted anomaly analysis to identify unusual patterns in the e-commerce dataset. • Analyzed and interpreted the identified anomalies using visual evidence. • Did report writing for Business Implications.
Wong Hui San (22034540)	• Conducted Literature Review. • Developed the Power BI dashboard, summary statistics, and visualizations. • Analyzed and interpreted the descriptive analytics results from Power BI. • Prepared the Results and Discussion section for the Power BI analysis. • Prepared the Conclusion and Future Improvement based on the overall project findings.

Appendix

Link to Power BI (pdf):
https://drive.google.com/file/d/1ZdjblE_PFTumq7EHq9qhO4S69peZiFrm/view?usp=sharing

Link to Google Colab (Data Preparation + ML Model Development):
https://colab.research.google.com/drive/1Bm_DO-Hc8lbSaaxzNRUPq2IbpecA7zaC?usp=sharing

Link to Google Colab (Anomaly Analysis):
<https://colab.research.google.com/drive/1zeW5OPnZ4rIECPY4vhi-RVVVAW590j6h?usp=sharing>